

Software Requirements Specification

Mudarabah-Enabled Distributed E-Procurement System

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This specification defines software requirements for a distributed e-procurement platform that automates procurement workflows and enables project-based mudarabah capital participation by banks, Islamic financial institutions, venture capital firms, or other approved financial entities. It is a tailored requirements artifact, not a legal contract, Shariah ruling, or production security assessment. Final use requires legal, Shariah, accounting, data protection, and regulatory review.

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1 Document Control and Tailoring Statement

1.1 Document identification

Title	Software Requirements Specification for the Mudarabah-Enabled Distributed E-Procurement System
Revision	Version 1.0, 2 June 2026
System of interest	Mudarabah-Enabled Procurement Network (MEPN)
Document type	Software Requirements Specification with UML appendices and traceability matrix
Intended readers	Product owner, software architects, developers, QA engineers, SME operators, financial entity stakeholders, Shariah/compliance reviewers, implementation partners

Table 1: Document identification

1.2 Tailoring to ISO/IEC/IEEE 29148

This SRS follows the 29148 information-item approach for a software requirements specification: purpose, scope, product overview, references, external interfaces, functional requirements, usability, performance, logical data, design constraints, software quality attributes, verification, assumptions, and acronyms. Additional stakeholder context, operational concept, UML models, and a traceability matrix are included because the product combines workflow automation, inter-organization collaboration, and regulated financing interactions.

The words **shall**, **should**, and **may** are used deliberately. Only statements using **shall** are binding requirements in this document. Descriptive, architectural, or rationale statements are non-binding unless explicitly written as requirements.

1.3 Source basis

The requirements are derived from the product vision in the prompt, the uploaded Malaysia mudarabah performance report, the uploaded procurement and mudarabah-procurement thesis documents, the uploaded ISO/IEC/IEEE 29148 standard, the referenced ERPNext repository, and official protocol references for OAuth/OpenID Connect and Hyperledger Fabric. The repository is used as an architectural reference point only; MEPN is specified as a rewrite with a different implementation stack.

2 Introduction

2.1 Purpose

The purpose of this SRS is to specify the required behavior, interfaces, data model, constraints, quality attributes, verification approach, and UML-supported analysis model for MEPN. The system is intended to make mudarabah financing more commercially adoptable by embedding financiers into high-integrity procurement workflows, reducing information asymmetry, improving project-level accounting evidence, and providing auditable cross-organization records.

2.2 Scope

MEPN is a distributed, self-hostable e-procurement and procurement-finance platform for SME organizations and financial entities. It supports conventional procurement automation, SME-

to-SME business interaction, ERP integration, supply-chain network visibility, and restricted project-based mudarabah financing workflows.

2.2.1 In scope

- Organization setup, user management, authorization, and role-based access control.
- Supplier onboarding, supplier master data, due diligence evidence, and supplier performance records.
- Source-to-contract functions: RFQ/RFP/tender publication, quotation management, evaluation, award, and contract record generation.
- Procure-to-pay functions: purchase requisition, approval, purchase order, goods receipt/service confirmation, invoice submission, three-way matching, payment status, and audit evidence.
- Procurement opportunity publication for revenue-generating trades, projects, resale cycles, or contract fulfillment.
- Mudarabah capital application, financier due diligence, Shariah/compliance review, contract execution, disbursement control, milestone monitoring, profit/loss calculation, profit distribution, and loss exception handling.
- Distributed deployment for SME-owned nodes with optional consortium participation through Hyperledger Fabric channels.
- Miro-like supply chain network canvas showing organizations, supplier/customer relationships, opportunities, financing links, document status, and risk indicators.
- ERP integration through APIs, file exchange, and adapter services.
- Audit trail, immutable event anchoring, evidence pack export, and reporting.

2.2.2 Out of scope for version 1.0

- Full banking core replacement, deposit account management, or capital adequacy computation.
- Automated legal enforceability determination or automated Shariah rulings without human reviewer approval.
- Public permissionless blockchain settlement or crypto-token issuance.
- General consumer purchases that do not generate separately measurable project profit.
- Autonomous credit approval without financier-defined policies and accountable human approval.
- Production regulatory reporting formats not yet specified by a target regulator.

2.3 Product perspective

MEPN is a distributed business application inspired by ERPNext's integrated ERP pattern. ERPNext exposes major ERP functions such as accounting, order management, manufacturing, asset management, and projects; the referenced repository also highlights a full-stack framework with database abstraction, user authentication, and REST API support, plus self-hosted Docker-oriented deployment. MEPN adopts the integration breadth and self-hostability pattern, but rewrites the system as a modular distributed procurement-finance platform.

2.4 Product functions

At a high level, MEPN provides the following product functions:

1. Automate procurement workflows from need identification to payment reconciliation.
2. Convert procurement documents into trusted financing evidence for project-based mudarabah.
3. Allow financial entities to review, approve, monitor, and close mudarabah-backed procurement ventures.
4. Provide distributed inter-organization visibility while preserving private business data.
5. Anchor critical inter-organization events on Hyperledger Fabric channels for tamper-evident auditability.
6. Integrate with SME ERP/accounting systems without requiring SMEs to surrender ownership of their deployment.
7. Visualize supply chain relationships and financing dependencies through a canvas-style network interface.

2.5 User characteristics

User class	Expected characteristics
SME Owner/Admin	Manages installation, users, organization profile, integration settings, and network participation. May have limited IT staff.
Procurement Officer	Creates requisitions, RFQs, POs, receipts, supplier records, and financing evidence packs. Familiar with P2P/S2C workflows.
Approver/Manager	Reviews business justification, budget, supplier selection, purchase orders, and financing requests. Needs fast decision support.
Supplier/Sales User	Responds to RFQs, acknowledges POs, submits delivery documents and invoices, and views payment status.
Finance/Accountant	Reconciles invoices, payments, project accounts, allowable costs, profit/loss calculations, and ERP postings.
Financial Entity Investment Officer	Evaluates opportunity quality, project economics, risk, and contract terms before capital commitment.
Shariah/Compliance Reviewer	Checks goods/services eligibility, contract form, profit ratio, loss treatment, documentation, and compliance exceptions.
Auditor	Reviews evidence, event logs, Fabric transaction hashes, user actions, changes, and contract closure packages.
Developer/Integrator	Builds adapters, chaincode integration, deployment automation, and custom extensions.

Table 2: User characteristics

2.6 Limitations

- Mudarabah commercial adoption depends on legal enforceability, financier appetite, Shariah governance, accounting discipline, and regulatory constraints that software can support but

not eliminate.

- Hyperledger Fabric is used for audit anchoring and inter-organization coordination, not as the full transactional database for confidential procurement records.
- Financial entities remain responsible for underwriting, suitability checks, liquidity treatment, contract approval, and regulatory obligations.
- SMEs remain responsible for accurate records, lawful procurement, tax compliance, delivery performance, and truthful profit/loss reporting.

2.7 Definitions

Mudarabah	A partnership in which a capital provider supplies capital and a manager/operator deploys effort and expertise. Profit is shared by pre-agreed ratio; genuine financial loss is borne by capital unless negligence, misconduct, fraud, or breach is proven.
Rabb-ul-Mal	Capital provider in a mudarabah arrangement; in this product it is usually a financial entity.
Mudarib	Entrepreneur or procurement operator that uses capital to fulfill a procurement opportunity.
Restricted mudarabah	A mudarabah arrangement restricted to a specified project, purpose, expense category, supplier set, buyer contract, and reporting regime.
P2P	Procure-to-pay: need, requisition, approval, purchase order, receipt, invoice verification, and payment.
S2C	Source-to-contract: requirement definition, sourcing, RFQ/RFP/tender, evaluation, award, and contract record.
Three-way match	Control that reconciles purchase order, receipt/service confirmation, and supplier invoice.
Network canvas	Visual graph of organizations, suppliers, buyers, financing links, documents, risks, and status flows.
Fabric channel	Permissioned Hyperledger Fabric administrative and ledger domain shared by selected organizations.

3 References

References

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- [4] *Inefficiencies in Traditional Procurement and the Performance Impact of Digital Procurement Solutions*, uploaded evidence review, May 2026.
- [5] raichiiiiiii/erpnext GitHub repository, default branch `develop`, README and source files inspected 2 June 2026.
- [6] RFC 6749, *The OAuth 2.0 Authorization Framework*, IETF/RFC Editor.
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- [8] Hyperledger Fabric Documentation, release 2.5, network, private data, MSP, and chaincode documentation.

4 Business and Stakeholder Context

4.1 Business problem

Traditional procurement frequently suffers from manual handoffs, fragmented data, weak controls, slow approvals, invoice exceptions, maverick spending, contract leakage, poor spend visibility, and limited auditability. These weaknesses also make procurement-backed mudarabah difficult because financiers require reliable borrower-level information, project accounting, cost verification, delivery evidence, and monitoring.

In Malaysia, the uploaded mudarabah performance report identifies a severe adoption gap for mudarabah financing relative to debt-like Islamic structures. The report attributes the gap to agency risk, profit verification cost, depositor expectations, regulatory/liquidity frictions, and weak borrower-level information. The product strategy is therefore to make procurement itself the evidence layer for mudarabah: verified buyer demand, supplier quotations, purchase orders, receipts, invoices, payments, and project ledgers become the operating data used to approve, monitor, and close financing.

4.2 Product vision

MEPN will be an installable, distributed e-procurement system for SMEs that automates procurement and embeds financial entity participation through restricted mudarabah contracts. Each SME can own its deployment, manage its organization data, connect its ERP, join selected supply-chain channels, and invite financiers into specific procurement opportunities without exposing unrelated business records.

4.3 Commercial adoption hypothesis

4.4 Stakeholders

4.5 Operational concept

An SME installs MEPN using a packaged deployment bundle. The SME configures its organization profile, roles, OAuth/OIDC identity provider, ERP adapter, supplier master data, and network policy. The SME uses standard procurement modules for requisitions, RFQs, POs, receipts, invoices, and payment status. When a procurement opportunity is revenue-generating and needs working capital, the SME creates a mudarabah capital application from verified procurement documents.

The financial entity accesses a restricted workspace for the opportunity. It reviews buyer credibility, supplier reliability, quotations, cost budget, delivery timeline, risk factors, Shariah eligibility, proposed profit ratio, and project account rules. If approved, the financial entity and SME execute a restricted mudarabah contract. Capital is disbursed either to a controlled

Mudarabah friction	Product response
Agency risk and moral hazard	Role-based workflows, controlled disbursement, document evidence, exception flags, and immutable audit events.
Profit verification cost	Project-specific ledger, allowable cost rules, PO/receipt/invoice/payment matching, ERP reconciliation, and financier verification workspace.
Weak borrower-level information	Procurement history, supplier performance, buyer contract evidence, delivery record, and financial event timeline.
Monitoring burden	Milestone dashboards, automated status alerts, Fabric event hashes, evidence pack export, and variance analytics.
Contract standardization gap	Configurable restricted mudarabah templates, versioned clauses, Shariah checklist, approval workflow, and contract lifecycle state model.
Liquidity and regulatory uncertainty	Product-level reporting exports, risk metrics, disbursement and realization schedules, and auditable closure data.

Table 3: Commercial adoption hypothesis for mudarabah support

account, directly to approved suppliers, or to the mudarib under defined controls. The system tracks procurement execution, delivery, invoice collection, project cost, profit/loss calculation, and distribution.

Critical cross-organization events are written to a Hyperledger Fabric channel as event hashes, transaction metadata, and optionally private data collection references. Full document payloads remain in the SME-controlled deployment unless the parties explicitly share them. The supply-chain canvas shows the buyer, supplier, financier, opportunity, documents, contract state, risks, and milestones.

4.6 System modes

5 Product Architecture Overview

5.1 Architecture style

MEPN is specified as a modular distributed application. The recommended implementation stack is:

- **Frontend:** React with TypeScript; graph/canvas layer such as React Flow; responsive web UI.
- **Backend:** TypeScript/NestJS modular services or equivalent clean architecture; REST/OpenAPI first, with event-driven internal boundaries.
- **Data:** PostgreSQL as the operational system of record, Redis for queues/cache/locks, object storage for documents, and OpenSearch-compatible search where needed.
- **Events:** NATS, RabbitMQ, or Kafka-compatible event bus for integration events, async retries, and audit propagation.
- **Identity:** OAuth 2.0 and OpenID Connect using external IdP integration or bundled identity provider such as Keycloak/ORY where permitted.
- **Distributed ledger:** Hyperledger Fabric Gateway integration for channel membership, endorsement, and tamper-evident event anchoring.

Stakeholder	Interest and influence
SME buyer organization	Uses procurement automation, supplier network, ERP integration, and audit records for internal purchasing.
SME supplier/mudarib	Uses procurement opportunities and project capital to fulfill buyer demand and generate profit.
Financial entity/rabb-ul-mal	Provides capital, reviews evidence, monitors venture performance, verifies profit/loss, and receives profit share where applicable.
Shariah/compliance function	Ensures goods/services, contract terms, profit ratio, loss treatment, reporting duties, and exceptions meet applicable standards.
Auditors and regulators	Require traceable evidence, immutable event records, access history, approval trails, and exportable reporting.
ERP/accounting system owner	Requires safe synchronization, master-data quality, idempotent posting, and reconciliation.
Implementation partner	Requires installable deployment, clear APIs, adapter points, migration tooling, and observability.
End customers or buyers	May supply purchase orders, accept delivery, approve invoices, and provide payment confirmation.

Table 4: Stakeholder summary

- **Deployment:** Docker Compose for SME installations, Helm/Kubernetes for managed or consortium deployments, and Infrastructure-as-Code templates for repeatability.

5.2 Logical architecture

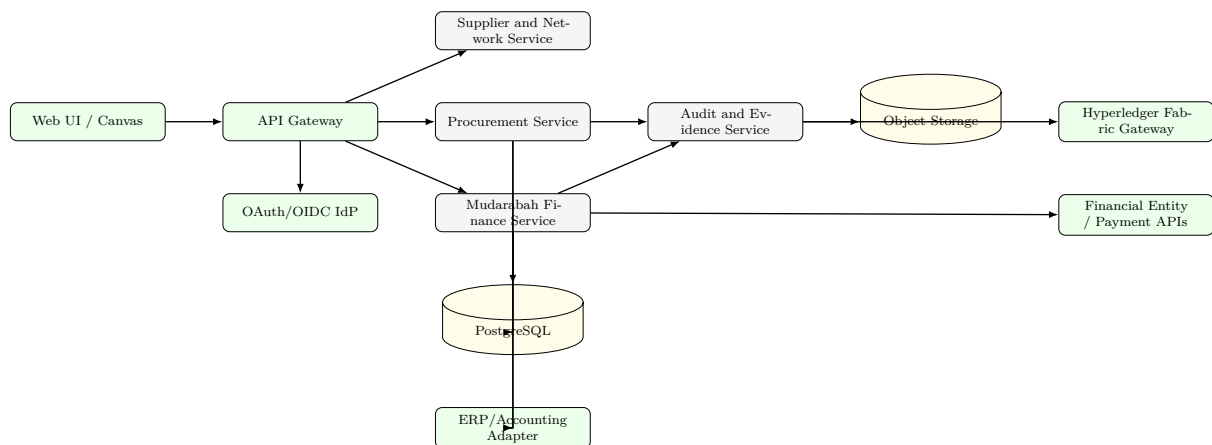


Figure 1: Logical architecture context for MEPN

5.3 Distributed deployment model

Each SME deployment is treated as an organization node with its own database, object store, identity configuration, audit log, and integration keys. Cross-organization collaboration uses signed API requests, Fabric event anchors, and invitation-based sharing. A financial entity may either install its own node, join as a Fabric organization, or access a hosted financier portal with contractual data segregation.

Mode	Description
Standalone SME mode	A single organization runs procurement workflows, ERP integration, and local audit logs without joining a Fabric consortium.
Federated procurement mode	Multiple SMEs exchange RFQs, quotations, POs, delivery records, and invoices across authenticated organization nodes.
Mudarabah finance mode	Financial entity users participate in a restricted workspace for application, due diligence, contract execution, monitoring, and closure.
Read-only audit mode	Auditors or reviewers access approved evidence and Fabric anchors without modifying business records.
Degraded integration mode	Core workflows continue when ERP, Fabric, or external payment APIs are unavailable; outbound events queue for retry.
Maintenance mode	Administrators perform backup, restore, migration, channel update, adapter reconfiguration, and security maintenance.

Table 5: System modes

5.4 Data confidentiality principle

The operational database stores full business records. Fabric stores event identifiers, hashes, timestamps, participating organizations, endorsement metadata, and optional private-data references. Unless an explicit sharing rule applies, confidential documents such as invoices, quotations, buyer contracts, and bank details remain off-chain in the owning deployment.

6 Specific Requirements

6.1 Requirement attributes

Each requirement is assigned an identifier, priority, verification method, and source. Priorities are: P0 mandatory for MVP viability, P1 mandatory for commercial beta, and P2 planned or extensibility requirement. Verification codes may combine inspection, analysis, demonstration, test, and review.

6.2 Stakeholder needs

ID	Priority	Requirement	Verification	Source
SN-01	High	SMEs need to automate procurement workflows without losing control of their own deployment and data.	Review	User vision, procurement thesis
SN-02	High	SME suppliers need access to working capital for confirmed revenue-generating procurement opportunities.	Review	Mudarabah-procurement thesis
SN-03	High	Financial entities need reliable procurement evidence to reduce agency risk and profit-verification cost.	Review	Mudarabah performance report
SN-04	High	Financial entities need controlled monitoring without taking over day-to-day procurement management.	Review	Mudarabah performance report
SN-05	High	Shariah and compliance reviewers need explicit checks for contract form, eligible activity, profit ratio, loss treatment, and breach clauses.	Review	Mudarabah-procurement thesis

ID	Priority	Requirement	Verification	Source
SN-06	Medium	Users need a visual supply-chain network view to understand counterparties, dependencies, risks, and financing links.	Demo	User vision
SN-07	High	Organizations need ERP/accounting integration to avoid duplicate records and support reliable project accounting.	Test	Procurement thesis, ERP template
SN-08	High	Auditors need tamper-evident records of key approvals, document versions, financing decisions, and contract lifecycle events.	Inspection	User vision, Fabric design
SN-09	Medium	Administrators need a packaged installation and maintenance workflow suitable for SMEs with limited IT resources.	Demo	User vision, ERP template
SN-10	Medium	Future features need an extension mechanism that does not compromise core procurement, finance, and audit integrity.	Inspection	User vision

6.3 Business requirements

ID	Priority	Requirement	Verification	Source
BR-01	P0	The system shall support an end-to-end procurement operating model covering source-to-contract and procure-to-pay activities.	T/D	SN-01
BR-02	P0	The system shall support restricted mudarabah financing for procurement opportunities that generate separately measurable revenue or margin.	T/D/R	SN-02,SN-03
BR-03	P0	The system shall maintain evidence sufficient to support due diligence, Shariah review, project monitoring, profit/loss calculation, and audit.	I/R	SN-03,SN-05,SN-08
BR-04	P0	The system shall prevent a mudarabah application from proceeding to contract execution until required procurement, financial, and Shariah evidence is complete or explicitly waived by authorized reviewers.	T/R	SN-03,SN-05
BR-05	P1	The system shall support SME-owned self-hosted deployment and optional managed deployment without changing the functional behavior of procurement and finance workflows.	T/I	SN-01,SN-09
BR-06	P1	The system shall permit an SME to invite selected counterparties and financial entities to a bounded opportunity workspace.	T/D	SN-02,SN-03
BR-07	P1	The system shall support tamper-evident event anchoring for cross-organization milestones and approvals.	T/I	SN-08
BR-08	P1	The system shall provide exportable evidence packs for financier review, Shariah review, dispute handling, and audit.	T/D	SN-03,SN-05,SN-08
BR-09	P1	The system shall provide visual supply-chain network visibility for authorized users.	D	SN-06
BR-10	P2	The system shall expose extension points for future features through versioned APIs, event subscriptions, and plugin interfaces.	I/T	SN-10

6.4 Functional requirements - identity, organization, and administration

ID	Priority	Requirement	Verification	Source
FR-01	P0	The system shall allow an organization administrator to create and maintain organization profile data, including legal name, registration number, tax identifiers, Shariah profile, contact points, and deployment mode.	T/D	BR-05

ID	Priority	Requirement	Verification	Source
FR-02	P0	The system shall support OAuth 2.0 authorization and OpenID Connect authentication for human users and service clients.	T/I	BR-05
FR-03	P0	The system shall enforce role-based access control for organization admin, procurement officer, approver, supplier user, finance user, financier user, Shariah reviewer, auditor, and developer/integrator roles.	T	BR-03
FR-04	P0	The system shall permit organization administrators to define approval matrices based on amount, category, project, department, risk level, and financing involvement.	T/D	BR-01
FR-05	P1	The system shall record every user login, permission change, organization invitation, and role assignment as an audit event.	T/I	BR-03, BR-07
FR-06	P1	The system shall support delegated access for financial entity reviewers limited to the approved opportunity workspace.	T	BR-06
FR-07	P1	The system shall provide API client credentials and signed webhook secrets for machine-to-machine integrations.	T/I	BR-10
FR-08	P2	The system shall support temporary emergency access grants with justification, expiry time, and mandatory audit review.	T/I	BR-03

6.5 Functional requirements - procurement core

ID	Priority	Requirement	Verification	Source
FR-09	P0	The system shall allow users to create purchase requisitions with item/service specification, quantity, delivery date, budget, department, project, and justification.	T/D	BR-01
FR-10	P0	The system shall route purchase requisitions through the configured approval matrix before procurement sourcing or purchase order creation.	T	BR-01
FR-11	P0	The system shall allow authorized users to create RFQ, RFP, tender, or direct quotation requests from approved requisitions.	T/D	BR-01
FR-12	P0	The system shall allow suppliers to submit quotations with price, delivery schedule, validity date, payment terms, compliance documents, and attachments.	T/D	BR-01
FR-13	P0	The system shall provide quotation comparison by price, delivery, supplier score, risk flags, Shariah eligibility, and total expected cost.	T/D	BR-01
FR-14	P0	The system shall allow authorized users to create purchase orders from approved quotations, contracts, or direct procurement approvals.	T/D	BR-01
FR-15	P0	The system shall track purchase order status from draft through submitted, acknowledged, partially received, received, billed, completed, cancelled, or closed.	T	BR-01
FR-16	P0	The system shall allow goods receipts and service confirmations to be recorded against purchase orders with quantity, quality, delivery evidence, and exception notes.	T/D	BR-01
FR-17	P0	The system shall allow supplier invoices to be matched against purchase orders and receipts or service confirmations.	T/D	BR-01
FR-18	P0	The system shall identify mismatches in quantity, price, tax, delivery, supplier bank details, or duplicate invoice reference before payment approval.	T	BR-01, BR-03

ID	Priority	Requirement	Verification	Source
FR-19	P1	The system shall maintain supplier onboarding records including registration, bank details, tax data, certifications, Shariah eligibility, risk documents, and approval status.	T/D	BR-01, BR-03
FR-20	P1	The system shall calculate supplier performance metrics from delivery timeliness, quality exceptions, invoice exceptions, response rate, and contract compliance.	T/A	BR-01
FR-21	P1	The system shall detect potential maverick spending by flagging purchases outside approved suppliers, catalogs, contracts, or approval rules.	T/A	BR-01
FR-22	P1	The system shall provide spend analytics by category, supplier, department, project, contract, financing involvement, and time period.	T/D	BR-01
FR-23	P1	The system shall allow procurement contracts and framework agreements to be stored with obligations, renewal dates, document versions, and linked POs.	T/D	BR-01
FR-24	P2	The system shall support configurable catalog procurement for repeat items and approved supplier price lists.	T/D	BR-01

6.6 Functional requirements - mudarabah finance workflow

ID	Priority	Requirement	Verification	Source
FR-25	P0	The system shall allow an authorized SME user to create a procurement opportunity from a buyer PO, contract award, tender result, sales order, or equivalent revenue-generating document.	T/D	BR-02
FR-26	P0	The system shall require each mudarabah opportunity to include expected revenue, cost budget, requested capital, delivery timeline, buyer identity, supplier plan, profit-sharing ratio, and risk assumptions.	T/I	BR-02, BR-03
FR-27	P0	The system shall prevent a mudarabah application from being submitted if the opportunity is marked as non-revenue-generating or routine internal consumption.	T	BR-02
FR-28	P0	The system shall generate a capital application evidence checklist based on opportunity type, sector, amount, buyer risk, supplier risk, and financier policy.	T/D	BR-03
FR-29	P0	The system shall provide a due diligence workspace where financial entity users can review buyer credibility, supplier reliability, cost reasonableness, delivery risk, accounting evidence, and repayment/profit realization assumptions.	T/D	BR-03
FR-30	P0	The system shall provide a Shariah/compliance review workspace covering eligible goods/services, contract restrictions, profit ratio, guaranteed return prohibition, loss treatment, negligence/breach clauses, and allowed expenses.	T/D/R	BR-03, BR-04
FR-31	P0	The system shall require explicit approval or rejection decisions from required financier and Shariah reviewers before contract generation.	T	BR-04
FR-32	P0	The system shall generate a restricted mudarabah contract record containing capital amount, project scope, profit ratio, disbursement conditions, reporting duties, permitted expenses, restrictions, loss treatment, breach clauses, dispute process, and signatures.	T/I	BR-02, BR-03
FR-33	P0	The system shall support electronic signature workflow or external e-signature integration for contract execution.	T/D	BR-02

ID	Priority	Requirement	Verification	Source
FR-34	P0	The system shall support disbursement controls including release to controlled account, release to approved supplier, milestone release, and manual release with justification.	T/D	BR-03
FR-35	P0	The system shall maintain a project-specific ledger of revenue, capital, procurement costs, allowed expenses, disbursements, buyer payments, profit, loss, and distribution.	T/A	BR-03
FR-36	P0	The system shall reconcile project costs against purchase orders, receipts, supplier invoices, payment records, and ERP accounting entries.	T/A	BR-03, BR-08
FR-37	P0	The system shall calculate preliminary profit or loss only after required revenue and allowable cost evidence is linked or waived by authorized reviewers.	T/A	BR-03
FR-38	P0	The system shall calculate profit distribution according to the approved profit-sharing ratio and shall not calculate a guaranteed fixed return on capital.	T/A/R	BR-02, BR-03
FR-39	P0	The system shall route loss cases to an exception workflow that distinguishes genuine commercial loss from negligence, misconduct, fraud, or breach evidence.	T/D/R	BR-03
FR-40	P1	The system shall provide financial entity dashboards for application pipeline, exposure, disbursement status, milestone risk, expected realization date, and closure status.	T/D	BR-03
FR-41	P1	The system shall allow a financial entity to define policy rules for sectors, maximum exposure, required documents, buyer eligibility, supplier restrictions, and Shariah review thresholds.	T/D	BR-03
FR-42	P1	The system shall export a mudarabah closure pack containing contract, approvals, procurement evidence, ledger, profit/loss calculation, distribution record, exceptions, and Fabric anchors.	T/I	BR-08

6.7 Functional requirements - network visibility, audit, and extension

ID	Priority	Requirement	Verification	Source
FR-43	P0	The system shall represent organizations, suppliers, buyers, financial entities, opportunities, contracts, documents, and financing links as a graph model.	T/I	BR-09
FR-44	P1	The system shall provide an interactive canvas where authorized users can add, move, group, filter, and annotate network nodes and edges.	T/D	BR-09
FR-45	P1	The system shall color or mark network nodes by status, risk, role, opportunity phase, financing phase, or document completeness.	D	BR-09
FR-46	P1	The system shall enforce access control on graph nodes and edges so that users see only relationships and metadata they are authorized to view.	T	BR-03, BR-09
FR-47	P0	The system shall create audit events for material cross-organization actions including invitations, RFQ publication, quotation submission, PO acknowledgement, receipt confirmation, invoice submission, financing decision, contract execution, disbursement, profit calculation, and closure.	T/I	BR-07
FR-48	P0	The system shall anchor selected audit events to Hyperledger Fabric by writing event hashes, organization identifiers, timestamps, and transaction references to approved channel chaincode.	T/I	BR-07

ID	Priority	Requirement	Verification	Source
FR-49	P1	The system shall queue Fabric anchoring events when the Fabric gateway is unavailable and retry with idempotency keys.	T	BR-07
FR-50	P1	The system shall allow authorized users to verify a document or event against its stored hash and Fabric transaction reference.	T/D	BR-07, BR-08
FR-51	P1	The system shall provide organization invitation and membership workflow for adding counterparties to a procurement workspace or network channel.	T/D	BR-06, BR-07
FR-52	P2	The system shall provide an extension registry for future modules and shall isolate extensions from core procurement, finance, and audit tables unless explicit permissions are granted.	I/T	BR-10

6.8 External interface requirements

ID	Priority	Requirement	Verification	Source
IR-01	P0	The system shall expose versioned REST APIs documented through OpenAPI for procurement, supplier, financing, audit, graph, and integration functions.	I/T	BR-10
IR-02	P0	The system shall support OAuth 2.0 authorization code with PKCE for browser-based user login when integrated with an external identity provider.	T/I	FR-02
IR-03	P0	The system shall validate ID token issuer, audience, expiry, signature, and nonce when OpenID Connect is used for authentication.	T/I	FR-02
IR-04	P0	The system shall support client credentials flow or signed JWT client authentication for trusted service integrations.	T/I	FR-07
IR-05	P0	The system shall support ERP synchronization for suppliers, items, purchase orders, receipts, invoices, payments, chart of accounts, and project ledger entries.	T/D	SN-07
IR-06	P0	The ERP adapter shall support idempotent create/update operations using external reference identifiers and reconciliation status.	T	SN-07
IR-07	P1	The system shall support CSV/XLSX import and export for SMEs that do not initially expose ERP APIs.	T/D	SN-07
IR-08	P0	The Fabric adapter shall connect to an approved gateway endpoint using organization identity material and channel configuration.	T/I	FR-48
IR-09	P0	The Fabric adapter shall submit transaction proposals to the configured chaincode and record resulting transaction IDs, block references, and endorsement status.	T/I	FR-48
IR-10	P1	The system shall support financial entity API integration for application submission, status retrieval, disbursement instruction, and payment confirmation where the financial entity provides APIs.	T/D	FR-29, FR-34
IR-11	P1	The system shall support external e-signature providers through a signed document package, callback validation, and final signed artifact storage.	T/D	FR-33
IR-12	P1	The system shall provide webhook subscriptions for procurement milestones, financing decisions, disbursement status, and audit events.	T/I	BR-10

6.9 Logical data requirements

ID	Priority	Requirement	Verification	Source
DR-01	P0	The logical data model shall include Organization, User, Role, Supplier, Buyer, Item, Project, ProcurementOpportunity, Requisition, RFQ, Quotation, PurchaseOrder, Receipt, Invoice, PaymentRecord, and AuditEvent.	I	BR-01
DR-02	P0	The logical data model shall include MudarabahApplication, DueDiligenceReport, ShariahReview, MudarabahContract, Disbursement, ProjectLedgerEntry, ProfitLossStatement, ProfitDistribution, and LossException.	I	BR-02, BR-03
DR-03	P0	Every material business object shall have organization ownership, lifecycle status, version, created/modified metadata, and audit correlation identifier.	I/T	BR-03
DR-04	P0	The system shall maintain immutable document versions after submission, approval, contract execution, or audit anchoring.	T/I	BR-03, BR-07
DR-05	P0	The system shall store only cryptographic hashes and minimum required metadata on Fabric unless private data collection rules explicitly permit additional data.	I/T	BR-07
DR-06	P1	The system shall support configurable retention policies by document type, organization, contract status, legal hold, and audit requirement.	T/I	BR-03
DR-07	P1	The system shall maintain a data lineage link from profit/loss calculations to source procurement documents, ERP postings, and reviewer decisions.	T/I	FR-35, FR-36
DR-08	P1	The system shall support master-data validation for supplier identity, bank account, tax details, item categories, units of measure, and organization identifiers.	T	FR-19
DR-09	P1	The system shall support anonymized or aggregated analytics exports that exclude confidential document payloads by default.	T/I	BR-09
DR-10	P2	The data model shall support custom attributes by organization without requiring core schema modification for common extension fields.	I/T	BR-10

6.10 Software system attributes and non-functional requirements

ID	Priority	Requirement	Verification	Source
NFR-01	P0	The system shall respond to 95 percent of interactive page requests within 2 seconds under the agreed SME reference load excluding external system latency.	Performance test	BR-01
NFR-02	P0	The system shall process 95 percent of standard procurement workflow transitions within 1 second after request validation excluding external integrations.	Performance test	BR-01
NFR-03	P1	The system shall support at least 100 concurrent users per SME deployment on the reference hardware profile defined in the deployment guide.	Load test	BR-05
NFR-04	P1	The system shall support at least 100,000 procurement documents per organization without functional degradation when indexed and archived according to guidance.	Load test	BR-05
NFR-05	P0	The system shall encrypt traffic using TLS 1.2 or higher for all browser, API, ERP adapter, financial entity, and Fabric gateway connections.	Security test	FR-02, IR-08
NFR-06	P0	The system shall encrypt secrets at rest using a managed secrets store or deployment-approved equivalent.	Inspection	FR-07
NFR-07	P0	The system shall enforce tenant and organization data isolation at the application service and database access layers.	Security test	BR-05

ID	Priority	Requirement	Verification	Source
NFR-08	P0	The system shall reject access tokens that are expired, issued by an untrusted issuer, signed by an untrusted key, or missing required audience/scope claims.	Security test	FR-02
NFR-09	P0	The system shall maintain an append-only audit log for authentication, authorization, workflow, data change, document, integration, and Fabric anchoring events.	Inspection/Test	BR-03, BR-07
NFR-10	P1	The system shall provide backup and restore procedures with a recovery point objective of 24 hours and a recovery time objective of 8 hours for SME self-hosted deployments.	DR test	BR-05
NFR-11	P1	The system shall queue outbound ERP, Fabric, e-signature, and financial entity events and retry them without duplicate business effects.	Integration test	IR-06, IR-09, IR-10
NFR-12	P1	The system shall continue core local procurement operations when Fabric is unavailable, with clear user indication that anchors are pending.	Resilience test	FR-49
NFR-13	P1	The system shall provide observable logs, metrics, traces, and health checks for application, database, queue, adapter, and Fabric connectivity components.	Inspection/Test	BR-05
NFR-14	P1	The system shall support configurable localization for currency, date format, tax labels, document numbering, and language packs.	Demo	BR-05
NFR-15	P1	The system shall meet WCAG 2.1 AA principles for core procurement and finance workflows where technically feasible.	Accessibility review	User characteristics
NFR-16	P1	The system shall provide clear validation messages for missing financing evidence, Shariah checklist failures, and procurement matching exceptions.	Usability test	BR-03, BR-04
NFR-17	P0	The system shall prevent modification of executed contract terms except through a versioned amendment workflow.	Security/Workflow test	FR-32
NFR-18	P0	The system shall not store full supplier bank account details in Fabric event payloads.	Inspection/Test	DR-05
NFR-19	P1	The system shall support deployment through Docker Compose with documented environment variables, secrets, migration, backup, and upgrade steps.	Installation test	BR-05
NFR-20	P2	The system should support Kubernetes deployment through Helm charts for financial entities or managed consortium operators.	Installation test	BR-05
NFR-21	P1	The system shall allow administrators to configure data residency boundaries for storage, backups, and Fabric channel participation.	Inspection/Demo	BR-05
NFR-22	P1	The system shall provide vulnerability scanning and dependency report generation as part of the release pipeline.	Inspection	Security
NFR-23	P1	The system shall support evidence export in PDF and machine-readable JSON package formats.	Test/Demo	BR-08
NFR-24	P2	The system should support plugin sandboxing to limit future extension access to explicit API scopes.	Security review	BR-10

6.11 Verification method definitions

Method	Definition
Inspection	Examination of documents, configuration, code, schema, logs, or deployed settings.
Analysis	Reasoned evaluation, calculation, model review, threat modeling, or consistency checking.
Demonstration	Observed execution of a workflow using representative data without deep instrumentation.

Method	Definition
Test	Repeatable procedure with expected input, output, assertions, and pass/fail criteria.
Review	Stakeholder, Shariah, legal, accounting, compliance, or operational acceptance review.

7 UML Requirements Model

7.1 Use Case Diagram

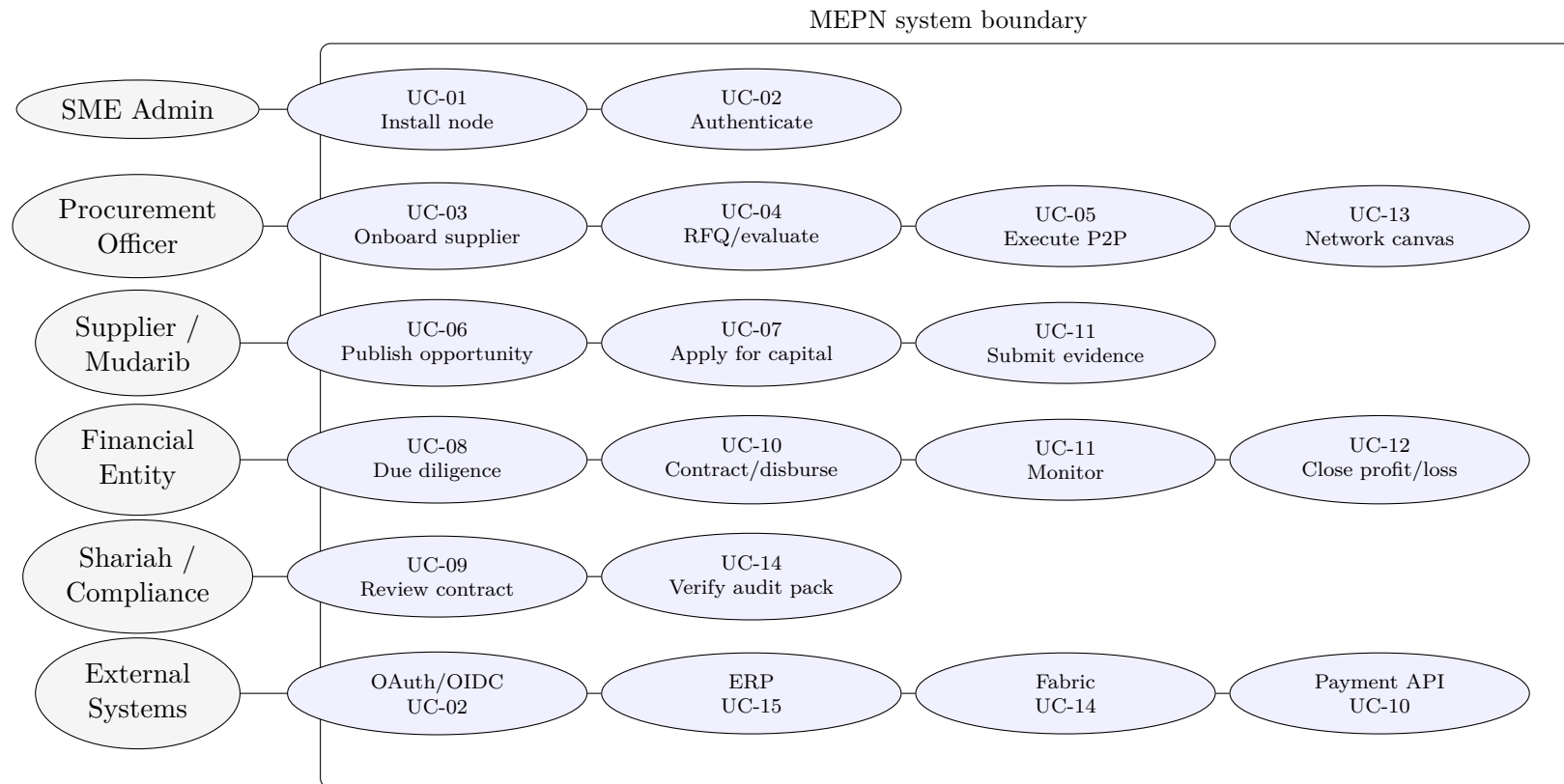


Figure 2: Use case diagram for MEPN showing actor-oriented use case groups

7.2 Use Case Specifications

7.2.1 UC-01 - Install and configure SME node

Use case ID and name	UC-01 - Install and configure SME node
Primary actor	SME Owner/Admin
Supporting actors	Developer/Integrator, Identity Provider, ERP system
Goal	Deploy a functioning self-hosted or managed organization node.
Preconditions	Installation package is available; administrator has infrastructure and organization details.
Trigger	Administrator starts first-run setup.
Main success scenario	1. Launch deployment. 2. Create organization profile. 3. Configure identity provider. 4. Configure database, object storage, email, and queue. 5. Create administrator role. 6. Run health checks. 7. Enable backup schedule.
Alternate or exception flows	A1: Identity provider unavailable - administrator uses local bootstrap admin and completes IdP configuration later. A2: ERP not ready - node operates without ERP adapter.
Postconditions	Organization node is operational, audited, and ready for procurement setup.
Related requirements	FR-01, FR-02, FR-03, NFR-10, NFR-19

7.2.2 UC-02 - Authenticate and authorize user

Use case ID and name	UC-02 - Authenticate and authorize user
Primary actor	Any human user
Supporting actors	OAuth/OIDC identity provider, API gateway
Goal	Access authorized system functions with correct organization context.
Preconditions	User exists in IdP or local invite workflow; organization role exists.
Trigger	User opens the web application.
Main success scenario	1. User starts login. 2. System redirects to IdP. 3. User authenticates. 4. System validates tokens. 5. System maps claims to organization roles. 6. Authorized landing page is shown.
Alternate or exception flows	A1: Invalid token - request is rejected. A2: Missing role - user sees no-access page and audit event is created.
Postconditions	Session is established with role-scoped permissions or access is denied.
Related requirements	FR-02, FR-03, FR-05, IR-02, IR-03, NFR-08

7.2.3 UC-03 - Onboard supplier

Use case ID and name	UC-03 - Onboard supplier
Primary actor	Procurement Officer
Supporting actors	Supplier User, Finance/Accountant, Shariah Reviewer
Goal	Create an approved supplier record with risk and compliance evidence.
Preconditions	Procurement officer has supplier management rights.
Trigger	A new supplier is required or a supplier responds to an invitation.
Main success scenario	1. Create supplier profile. 2. Request documents. 3. Supplier submits registration, bank, tax, certification, and Shariah information. 4. System validates completeness. 5. Reviewer approves, rejects, or requests changes. 6. Supplier becomes available for sourcing.
Alternate or exception flows	A1: Bank details mismatch - supplier is blocked for payment use. A2: Shariah eligibility unresolved - supplier can be marked restricted.
Postconditions	Supplier has status, evidence, and approval history.
Related requirements	FR-19, FR-20, DR-08, NFR-16

7.2.4 UC-04 - Run RFQ and supplier evaluation

Use case ID and name	UC-04 - Run RFQ and supplier evaluation
Primary actor	Procurement Officer
Supporting actors	Supplier User, Approver/Manager
Goal	Select a supplier through controlled sourcing.
Preconditions	Approved requisition or sourcing need exists.
Trigger	Procurement officer creates RFQ/RFP/tender.
Main success scenario	1. Create RFQ from requirement. 2. Invite suppliers. 3. Suppliers submit quotations. 4. System compares quotations. 5. Procurement officer recommends award. 6. Approver approves. 7. Award record is stored.
Alternate or exception flows	A1: Supplier misses deadline - quotation is excluded unless authorized exception. A2: Conflict of interest flag - additional approval required.
Postconditions	Approved supplier quotation or award exists for PO creation and optional financing evidence.
Related requirements	FR-11, FR-12, FR-13, FR-23

7.2.5 UC-05 - Execute procure-to-pay workflow

Use case ID and name	UC-05 - Execute procure-to-pay workflow
Primary actor	Procurement Officer
Supporting actors	Approver/Manager, Supplier User, Finance/Accountant, ERP system
Goal	Create and close a purchase transaction with matching evidence.
Preconditions	Approved requisition or quotation exists; supplier is approved.
Trigger	Purchase order is created.
Main success scenario	1. Generate PO. 2. Route for approval. 3. Supplier acknowledges. 4. Goods or services are received. 5. Supplier submits invoice. 6. System performs three-way match. 7. Finance approves payment status or ERP posting. 8. PO is completed or closed.
Alternate or exception flows	A1: Match exception - invoice enters exception workflow. A2: Partial delivery - PO remains partially received. A3: Supplier on hold - PO submission blocked.
Postconditions	Procurement transaction has auditable document chain and ERP reconciliation status.
Related requirements	FR-09 to FR-18, IR-05, IR-06

7.2.6 UC-06 - Publish procurement opportunity for financing

Use case ID and name	UC-06 - Publish procurement opportunity for financing
Primary actor	SME Supplier/Mudarib
Supporting actors	Buyer, Procurement Officer, Finance/Accountant
Goal	Create a revenue-generating procurement opportunity that can be financed.
Preconditions	Buyer PO, sales order, tender award, or contract evidence exists.
Trigger	Mudarib needs capital to execute the opportunity.
Main success scenario	1. Create opportunity record. 2. Attach buyer demand evidence. 3. Add expected revenue, supplier plan, cost budget, delivery timeline, risk assumptions, and requested capital. 4. System checks suitability. 5. Opportunity is submitted for financing.
Alternate or exception flows	A1: Opportunity is internal consumption - system blocks mudarabah submission. A2: Evidence missing - application remains draft.
Postconditions	Opportunity is ready for financier invitation or internal review.
Related requirements	FR-25, FR-26, FR-27, FR-28

7.2.7 UC-07 - Apply for mudarabah capital

Use case ID and name	UC-07 - Apply for mudarabah capital
Primary actor	SME Supplier/Mudarib
Supporting actors	Financial Entity User, Shariah Reviewer

Goal	Submit a complete capital application for restricted mudarabah financing.
Preconditions	Procurement opportunity exists and passes suitability checks.
Trigger	Mudarib sends application to a financial entity.
Main success scenario	1. Select financier. 2. Confirm requested capital and profit ratio proposal. 3. Submit evidence checklist. 4. System creates restricted workspace. 5. Financial entity receives application notification.
Alternate or exception flows	A1: Financier policy requires extra documents - system requests additional evidence. A2: Profit ratio omitted - submission blocked.
Postconditions	Application is in submitted state with audit record and workspace access.
Related requirements	FR-26, FR-28, FR-29, FR-41

7.2.8 UC-08 - Perform financier due diligence

Use case ID and name	UC-08 - Perform financier due diligence
Primary actor	Financial Entity Investment Officer
Supporting actors	SME Supplier/Mudarib, Buyer, Supplier, ERP system
Goal	Assess commercial viability and risk of the application.
Preconditions	Application is submitted and reviewer has workspace access.
Trigger	Investment officer opens due diligence task.
Main success scenario	1. Review buyer evidence. 2. Review supplier plan and quotations. 3. Review project economics. 4. Review ERP/accounting history. 5. Record findings and conditions. 6. Approve, reject, or request changes.
Alternate or exception flows	A1: Buyer cannot be verified - application is rejected or paused. A2: Cost budget unreasonable - application returns for revision.
Postconditions	Due diligence decision is recorded with evidence and conditions.
Related requirements	FR-29, FR-41, DR-07

7.2.9 UC-09 - Perform Shariah and compliance review

Use case ID and name	UC-09 - Perform Shariah and compliance review
Primary actor	Shariah/Compliance Reviewer
Supporting actors	Financial Entity User, SME Supplier/Mudarib
Goal	Validate contract eligibility and compliance controls.
Preconditions	Application and due diligence evidence are available.
Trigger	Reviewer receives Shariah/compliance review task.
Main success scenario	1. Review goods/services. 2. Review buyer and supplier restrictions. 3. Review profit ratio. 4. Review loss and negligence clauses. 5. Review allowed expenses. 6. Approve, reject, or request amendments.
Alternate or exception flows	A1: Goods/services not eligible - application rejected. A2: Profit term implies guaranteed fixed return - amendment required.
Postconditions	Compliance decision and checklist are stored.
Related requirements	FR-30, FR-31, FR-32

7.2.10 UC-10 - Execute mudarabah contract and disburse capital

Use case ID and name	UC-10 - Execute mudarabah contract and disburse capital
Primary actor	Financial Entity User
Supporting actors	SME Supplier/Mudarib, Shariah Reviewer, E-signature provider, Payment API
Goal	Create binding contract record and release capital under approved controls.
Preconditions	Due diligence and Shariah review are approved.
Trigger	Financial entity initiates contract execution.

Main success scenario	1. Generate contract from approved terms. 2. Parties review. 3. Parties sign. 4. System locks contract version. 5. Financial entity chooses disbursement method. 6. Disbursement is recorded and optionally sent to payment API. 7. Audit/Fabric event is created.
Alternate or exception flows	A1: Signature rejected - contract returns for correction. A2: Payment API unavailable - disbursement instruction is queued or manually recorded.
Postconditions	Contract is executed and capital release status is recorded.
Related requirements	FR-32, FR-33, FR-34, FR-47, FR-48

7.2.11 UC-11 - Monitor procurement execution

Use case ID and name	UC-11 - Monitor procurement execution
Primary actor	Financial Entity User
Supporting actors	SME Supplier/Mudarib, Procurement Officer, Supplier User, Buyer
Goal	Track milestones without interfering with mudarib management.
Preconditions	Contract executed and project ledger active.
Trigger	Procurement execution begins or milestone changes.
Main success scenario	1. System imports or records POs, receipts, invoices, payments, and exceptions. 2. Dashboard updates milestone and risk status. 3. User reviews variances. 4. Evidence requests are issued if needed. 5. Critical events are anchored.
Alternate or exception flows	A1: Delivery delay - risk flag and notification. A2: Cost overrun - approval or amendment workflow. A3: Missing receipt - profit calculation blocked.
Postconditions	Project status and evidence are current.
Related requirements	FR-35, FR-36, FR-40, FR-47, FR-49

7.2.12 UC-12 - Calculate profit/loss and close mudarabah project

Use case ID and name	UC-12 - Calculate profit/loss and close mudarabah project
Primary actor	Finance/Accountant
Supporting actors	Financial Entity User, SME Supplier/Mudarib, Shariah Reviewer, Auditor
Goal	Compute and approve project outcome and distribution.
Preconditions	Revenue, cost, payment, and expense evidence is complete or waived.
Trigger	Buyer payment is received or project closure is requested.
Main success scenario	1. System compiles project ledger. 2. System calculates allowable costs. 3. System calculates net profit or loss. 4. Reviewers verify calculation. 5. Profit is distributed by approved ratio or loss workflow is started. 6. Closure pack is generated. 7. Contract is closed.
Alternate or exception flows	A1: Loss detected - loss exception workflow determines genuine loss vs breach. A2: Expense disputed - calculation paused pending resolution.
Postconditions	Project is closed with auditable calculation and distribution or loss decision.
Related requirements	FR-35, FR-37, FR-38, FR-39, FR-42

7.2.13 UC-13 - Use supply-chain network canvas

Use case ID and name	UC-13 - Use supply-chain network canvas
Primary actor	Procurement Officer or Financial Entity User
Supporting actors	Graph service, audit service
Goal	Visualize relationships, status, risk, and finance dependencies.
Preconditions	User has graph view permission.
Trigger	User opens network canvas.
Main success scenario	1. System loads authorized graph nodes and edges. 2. User filters by organization, supplier, opportunity, status, risk, or financing phase. 3. User expands node details. 4. User adds comments or annotations. 5. System saves view snapshot.

Alternate or exception flows	A1: User lacks permission - node metadata is redacted. A2: Graph query is large - system prompts filter refinement.
Postconditions	User obtains authorized network visibility without exposing unauthorized data.
Related requirements	FR-43, FR-44, FR-45, FR-46

7.2.14 UC-14 - Verify audit event and evidence pack

Use case ID and name	UC-14 - Verify audit event and evidence pack
Primary actor	Auditor
Supporting actors	Fabric Gateway, Audit Service, Object Storage
Goal	Confirm that a document or event matches stored system and Fabric evidence.
Preconditions	Auditor has read-only access to evidence pack.
Trigger	Audit review begins or dispute arises.
Main success scenario	1. Auditor opens evidence pack. 2. System shows document version, hash, workflow history, user actions, and Fabric transaction reference. 3. Auditor verifies hash. 4. System retrieves Fabric transaction metadata. 5. Auditor records finding.
Alternate or exception flows	A1: Fabric unavailable - local evidence is shown and chain verification marked pending. A2: Hash mismatch - exception is raised.
Postconditions	Audit verification status is recorded.
Related requirements	FR-47, FR-48, FR-50, FR-42

7.2.15 UC-15 - Integrate ERP/accounting records

Use case ID and name	UC-15 - Integrate ERP/accounting records
Primary actor	Developer/Integrator
Supporting actors	ERP system, Finance/Accountant, API Gateway
Goal	Synchronize procurement and project-accounting records between systems.
Preconditions	ERP adapter is configured and credentials are approved.
Trigger	A procurement or finance event requires external posting or import.
Main success scenario	1. Adapter maps master data. 2. System sends or receives document event. 3. Adapter validates idempotency key. 4. ERP confirms result. 5. System records reconciliation status. 6. Exceptions are queued for correction.
Alternate or exception flows	A1: ERP rejects posting - exception task is created. A2: Duplicate event - adapter returns existing result.
Postconditions	Records are synchronized or exception is visible.
Related requirements	IR-05, IR-06, FR-36, NFR-11

7.3 Activity Diagrams for Major Workflows

7.3.1 Activity A1 - Source-to-contract and procure-to-pay

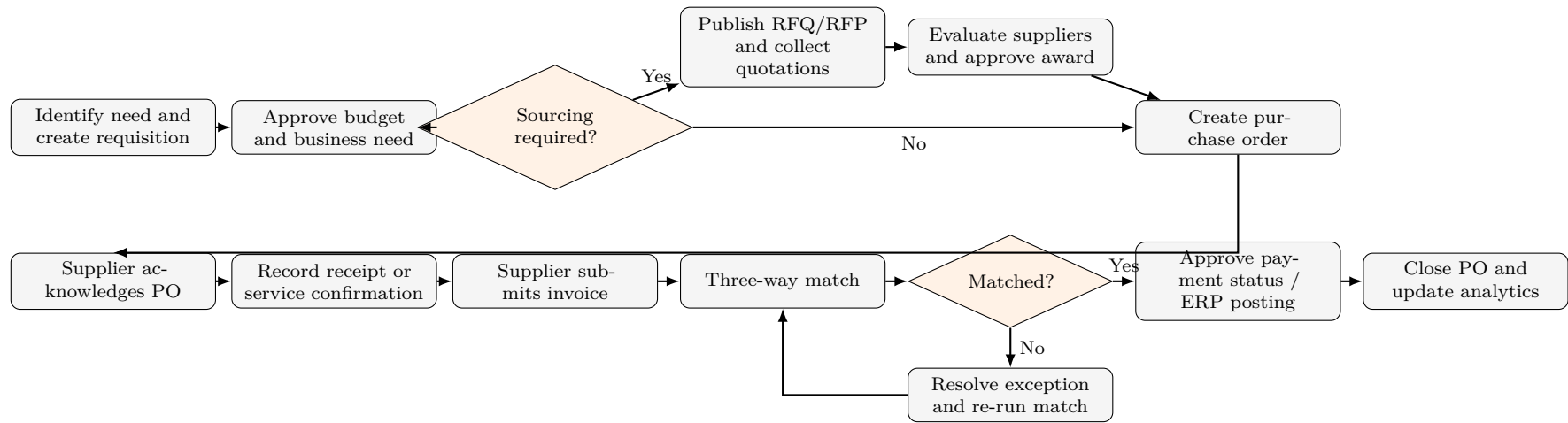


Figure 3: Activity diagram - procurement workflow

7.3.2 Activity A2 - Mudarabah capital application and approval

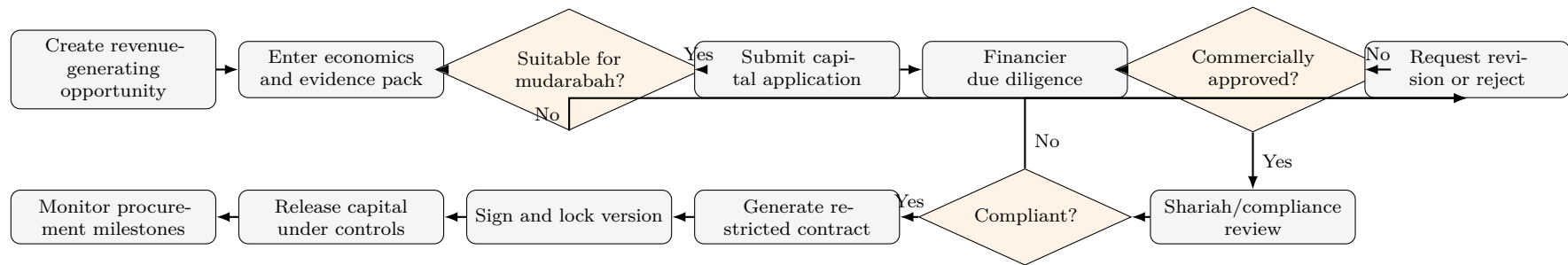


Figure 4: Activity diagram - mudarabah application and approval

7.3.3 Activity A3 - Project monitoring, profit/loss, and closure

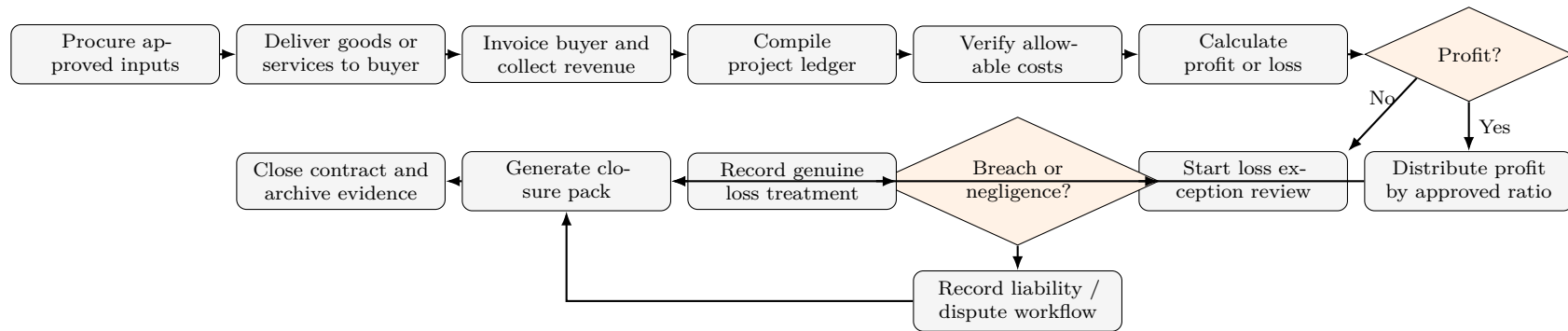


Figure 5: Activity diagram - mudarabah monitoring and closure

7.3.4 Activity A4 - Supply-chain network canvas and audit anchoring

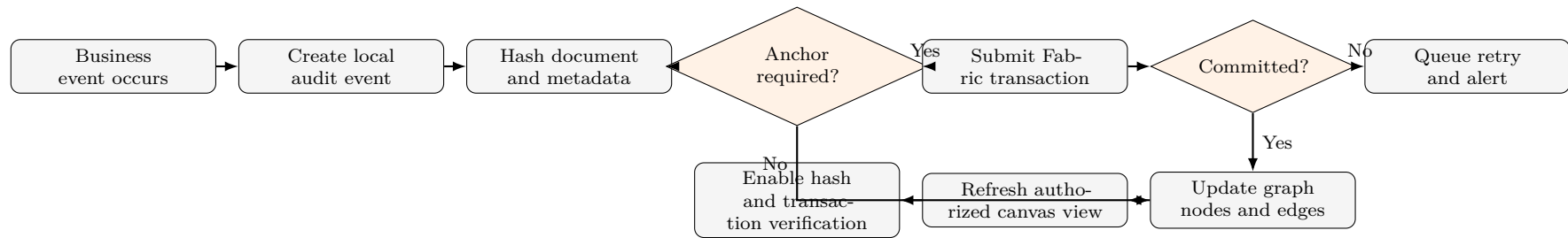


Figure 6: Activity diagram - network canvas and audit anchoring

7.4 Domain Class Diagram

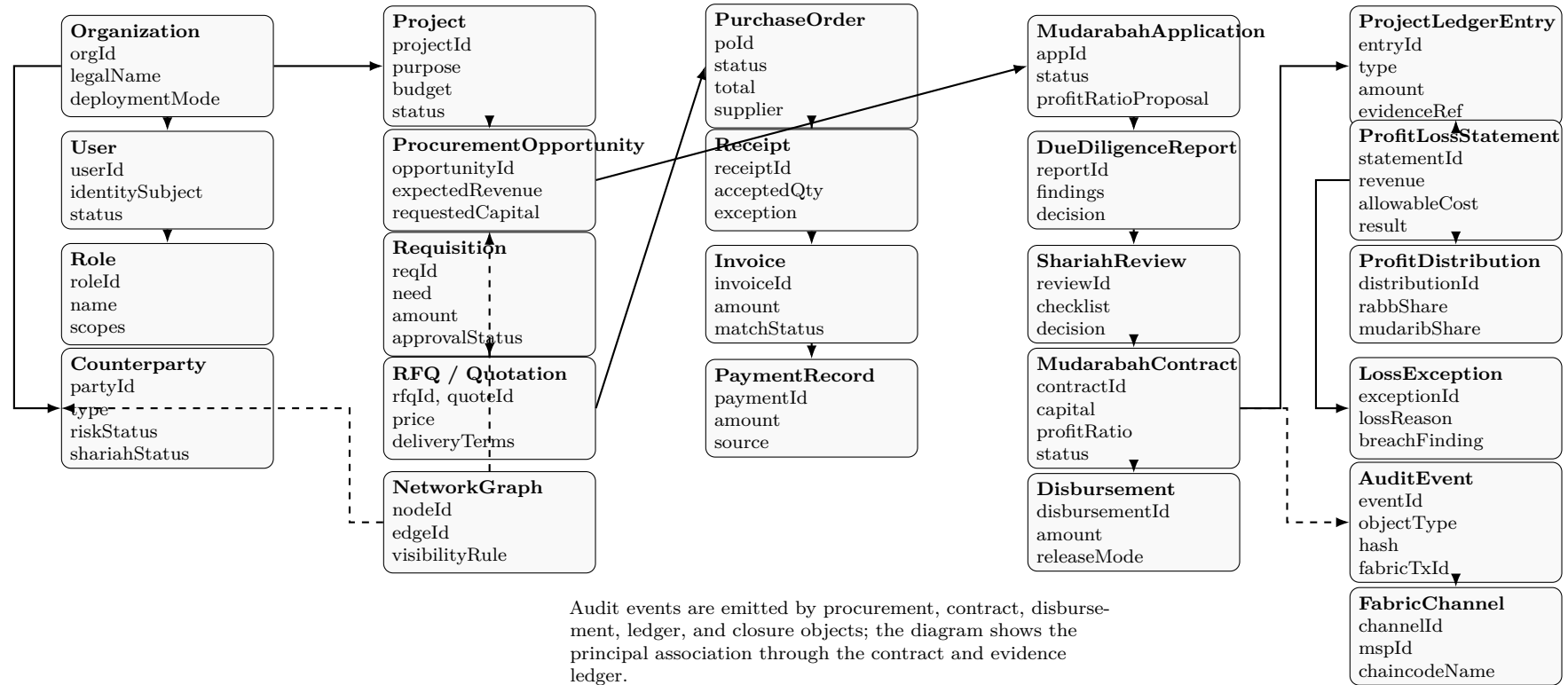


Figure 7: Domain class diagram showing principal classes and associations

7.5 State Machine Diagram - Mudarabah Contract Lifecycle

The mudarabah contract lifecycle matters because capital release, loss treatment, auditability, and Shariah/compliance controls depend on state transitions. The diagram applies to the combined application-contract lifecycle.

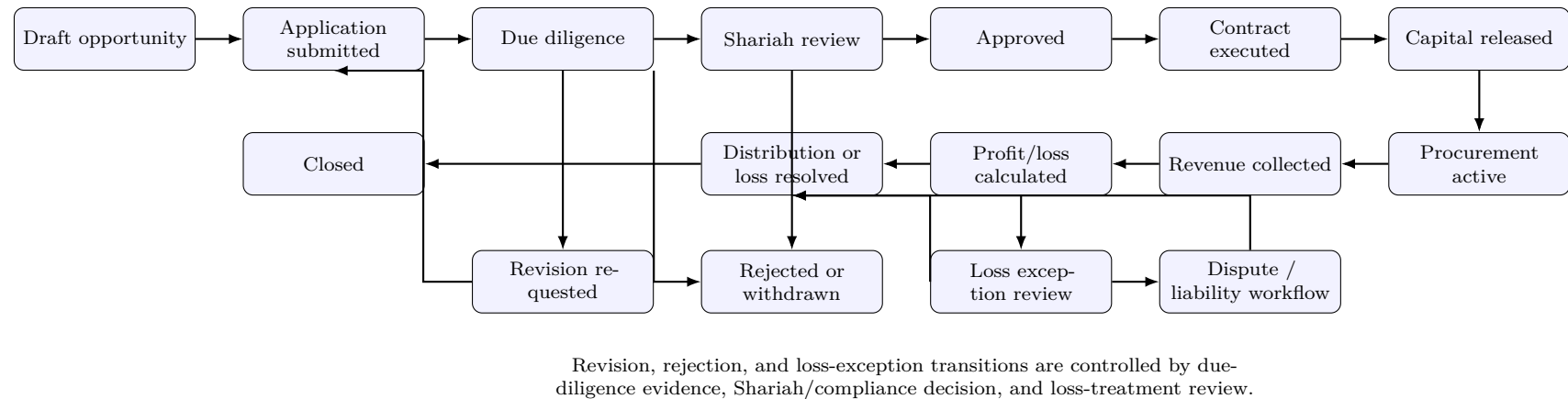


Figure 8: State machine - mudarabah application and contract lifecycle

7.6 Sequence Diagrams for External Integrations

Sequence diagrams are included only where MEPN interacts with external systems.

7.6.1 Sequence S1 - OAuth/OIDC login

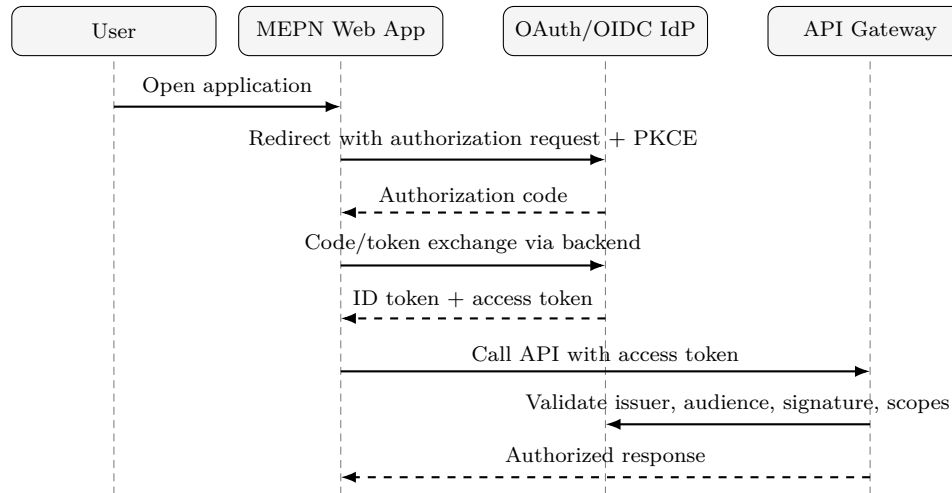


Figure 9: Sequence diagram - OAuth/OIDC login

7.6.2 Sequence S2 - ERP synchronization

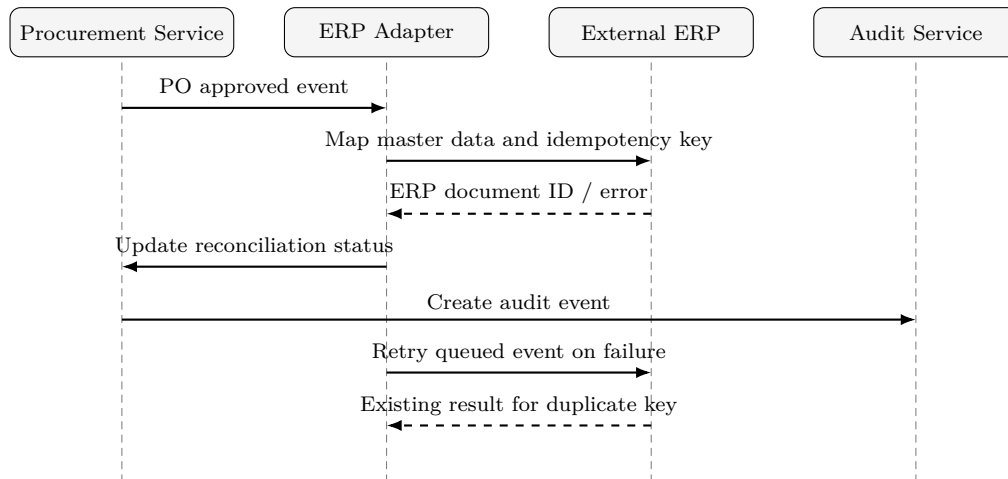


Figure 10: Sequence diagram - external ERP integration

7.6.3 Sequence S3 - Hyperledger Fabric event anchoring

7.6.4 Sequence S4 - Financial entity disbursement API

8 Verification Plan

8.1 Verification strategy

Verification shall be organized by requirement class and release stage. P0 requirements shall be verified before MVP acceptance. P1 requirements shall be verified before commercial beta. P2

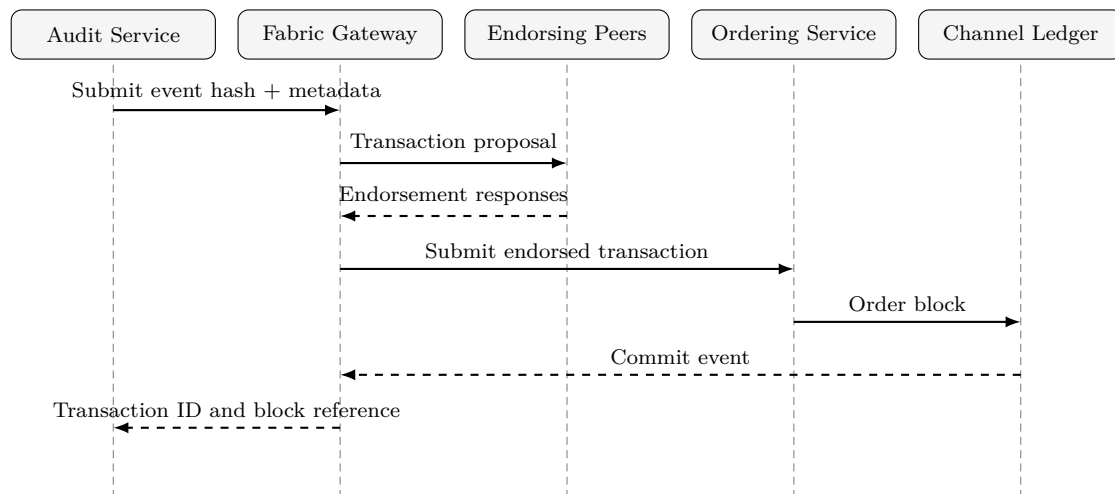


Figure 11: Sequence diagram - Fabric event anchoring

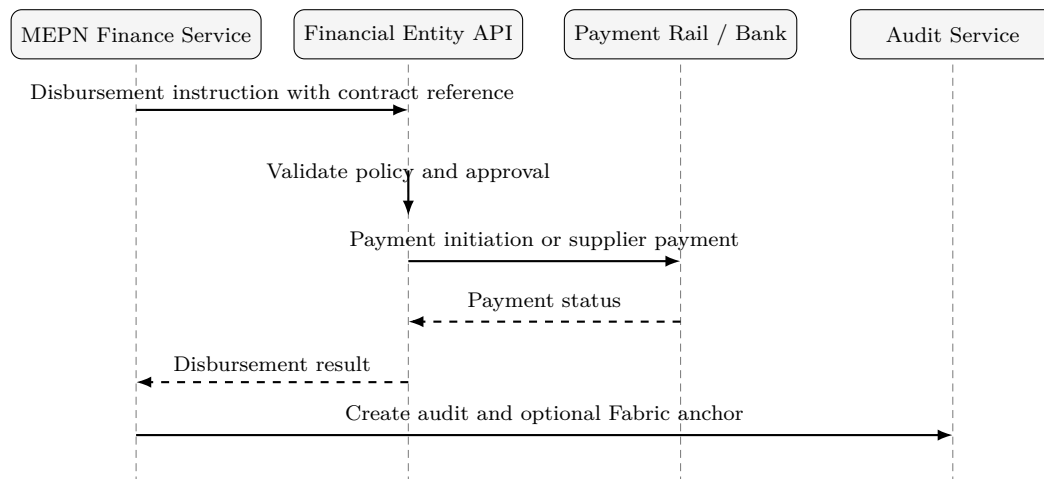


Figure 12: Sequence diagram - external financial entity disbursement API

requirements shall be verified before optional release activation. Every verification record shall link to the requirement ID, test artifact, tester/reviewer, date, input data, result, evidence, and defect reference where applicable.

8.2 Verification coverage by requirement group

8.3 Acceptance criteria for MVP

The MVP shall not be accepted unless all P0 requirements are either passed or formally waived by the product owner, system owner, and risk owner. Waivers for Shariah, legal, security, or financial control requirements shall require documented reviewer sign-off and expiry date.

9 Traceability Matrix

Requirement group	Minimum verification evidence
Identity and authorization	Token validation tests, role matrix tests, unauthorized access attempts, audit log inspection.
Procurement workflow	Scenario tests for requisition, RFQ, quotation, PO, receipt, invoice, match exception, payment status, and closure.
Mudarabah finance	Scenario tests for application completeness, due diligence, Shariah review, contract generation, disbursement, project ledger, profit/loss, and loss exception.
ERP integration	Adapter mapping tests, idempotency tests, duplicate handling, reconciliation status, and failure retry tests.
Fabric integration	Chaincode transaction tests, event hash verification, failed gateway retry, transaction ID recording, and channel access control.
Network canvas	Graph permission tests, filter/search tests, redaction tests, node/edge update tests, and snapshot export tests.
Security and audit	Penetration-oriented test cases, secrets inspection, TLS checks, immutable audit log checks, and contract amendment workflow tests.
Deployment and operations	Installation test, migration test, backup/restore test, health check, observability review, and upgrade rollback rehearsal.

Table 31: Verification coverage summary

Need	Need summary	Business reqs	Software reqs	UML coverage	Verification evidence
SN-01	SME-controlled procurement	e- BR-01, BR-05	FR-01 to FR-24, IR-05 to IR-07, NFR-19	UC-01, UC-04, UC-05, UC-15	Test, demonstration, install test
SN-02	Working capital for procurement opportunities	BR-02, BR-06	FR-25 to FR-28, FR-34	UC-06, UC-07, UC-10	Scenario test, review
SN-03	Reduce financier information asymmetry	BR-03, BR-04, BR-08	FR-28 to FR-42, DR-02, DR-07	UC-07 to UC-12	Due diligence review, evidence pack inspection
SN-04	Monitoring without management takeover	BR-03, BR-07	FR-35, FR-36, FR-40, FR-47 to FR-50	UC-11, UC-14	Dashboard demo, audit verification
SN-05	Shariah and compliance governance	BR-03, BR-04	FR-30 to FR-32, FR-38, FR-39	UC-09, UC-12	Reviewer acceptance, workflow test
SN-06	Supply-chain network visibility	BR-09	FR-43 to FR-46, DR-09	UC-13	Canvas demo, permission test
SN-07	ERP/accounting integration	BR-01, BR-03	IR-05 to IR-07, FR-36, DR-07, NFR-11	UC-05, UC-15	Integration test, reconciliation test
SN-08	Tamper-evident auditability	BR-03, BR-07, BR-08	FR-47 to FR-50, DR-04, DR-05, NFR-09	UC-10, UC-11, UC-14	Fabric test, audit inspection
SN-09	Installable SME deployment	BR-05	FR-01, NFR-10, NFR-13, NFR-19, NFR-20	UC-01	Install, backup, restore, health check
SN-10	Future feature extension	BR-10	FR-52, IR-01, IR-12, DR-10, NFR-24	UC-01, UC-15	API inspection, plugin sandbox review

A Assumptions and Dependencies

A.1 Assumptions

1. SMEs will operate the system for revenue-generating procurement opportunities where project-level profit/loss can be measured.
2. Financial entities will define underwriting policies, reviewer roles, product limits, disbursement controls, and Shariah requirements.
3. At least one deployment mode will be available for SMEs with limited IT resources through Docker Compose.
4. Fabric channel governance will be agreed by participating organizations before production use.
5. ERP integration will vary by SME maturity; CSV/XLSX exchange is therefore retained as a fallback.
6. Legal enforceability, Shariah approval, accounting treatment, and regulatory reporting requirements will be finalized per jurisdiction and institution.

A.2 Dependencies

- OAuth/OIDC identity provider or bundled identity component.
- PostgreSQL, object storage, queue/cache, email provider, and backup storage.
- Hyperledger Fabric gateway, channel configuration, MSP identity material, and approved chaincode for audit anchors.
- ERP/accounting APIs or import/export capabilities.
- Optional e-signature provider, payment rail, banking API, and financial entity workflow API.
- Legal, Shariah, accounting, cybersecurity, and data protection review before production release.

B Acronyms and Abbreviations

Acronym	Meaning
API	Application Programming Interface
ERP	Enterprise Resource Planning
IdP	Identity Provider
MEPN	Mudarabah-Enabled Procurement Network
MSP	Membership Service Provider in Hyperledger Fabric
OIDC	OpenID Connect
P2P	Procure-to-pay
PO	Purchase Order
RBAC	Role-Based Access Control
RFQ/RFP/RFI	Request for Quotation / Proposal / Information
RTM	Requirements Traceability Matrix
S2C	Source-to-contract
SME	Small and Medium Enterprise

SRS	Software Requirements Specification
TLS	Transport Layer Security
UML	Unified Modeling Language

C Open Issues for Product Discovery

ID	Issue	Owner
OI-01	Select the first target jurisdiction and map legal/Shariah/regulatory requirements to production controls.	Product + Legal/Shariah
OI-02	Decide whether the financial entity portal is always self-hosted by the financier or can be hosted by an SME consortium operator.	Product Architecture
OI-03	Define minimum borrower accounting maturity required for live mudarabah financing.	Finance/Risk
OI-04	Select initial ERP adapters and document canonical field mappings.	Integration Team
OI-05	Define Fabric channel topology, endorsement policies, and whether private data collections are required for first release.	Distributed Ledger Team
OI-06	Prioritize future discovered features such as AI-assisted risk scoring, anomaly detection, negotiation support, or automated document extraction.	Product Owner

D Conformance Notes

This document is intended as a requirements specification artifact aligned with ISO/IEC/IEEE 29148 content guidance. It does not assert that the complete project lifecycle, organizational process, or all information items have achieved full standard conformance. It establishes a requirements baseline that can be reviewed, changed, and traced through formal requirements management.